

Thanks to this technique, we have been able to reveal that the hidden symmetry algebra emerging from the embedding chain $\mathfrak{su}(4) \supset \mathfrak{su}(2) \times \mathfrak{su}(2)$ gives rise to a quartic polynomial Poisson algebra. As a byproduct, this paper has also demonstrated that the grading method not only simplifies complex expansions in the Poisson bracket relations of a given degree but is, in fact, indispensable for deriving these relations when the degree becomes so high that standard “brute-force” methods are no longer applicable.

This ansatz can be generalized and applied to any other reduction chain $\mathfrak{g} \supset \mathfrak{g}'$ involving reductive Lie algebras where the subalgebra embedding is not regular, and for which the usual argumentation using the root system is no longer applicable [10]. However, as shown in [39], there are several alternatives to define non-canonical gradings in Lie algebras, that can be conveniently adapted to various non-regular reduction chains (see [40, 41] and references therein). Following this argumentation, the reduction chains of interest in physical applications (see e.g. [31, Chapter 12], [42]) examined in the context of the missing label problem can be reevaluated from the perspective of polynomial algebras, allowing us to describe the complete algebraic structure, and potentially allowing us to make more effective of labeling operators to separate degenerate states. For applications in superintegrable systems [43, 44], the precise knowledge of the associated polynomial algebra can be useful for the systematic construction of Hamiltonians admitting constants of the motion of degrees higher than two, providing new hierarchies of systems for which the classical criteria for the separation of variables no longer necessarily apply [45, 46]. It is worthy to be mentioned in this context that the observed relations between polynomial algebras and quasi-exactly solvable problems (see e.g. [47]) could be systematized by means of a detailed analysis of the (hidden) symmetry algebras and their associated polynomial structures. Work in these directions is currently in progress.

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